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$$|z| = \sqrt{Re(z)^2 + Im(z)^2} = \sqrt{a^2 + b^2}$$

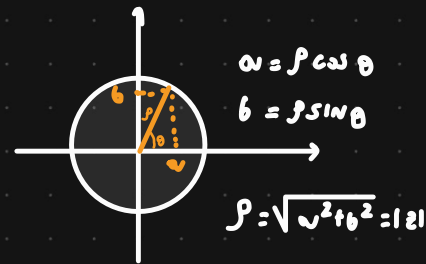
$$z = a + ib$$

$$|z|^2 = z \cdot z^* = (a+ib)(a-ib)$$

EULER'S FORMULA

$$e^{i\theta} = \cos\theta + i\sin\theta \quad e^{i\pi} + 1 = 0$$

$$z = \rho e^{i\theta} = \rho(\cos\theta + i\sin\theta)$$



$$z_1 = \rho_1 e^{i\theta_1}$$

$$z_2 = \rho_2 e^{i\theta_2}$$

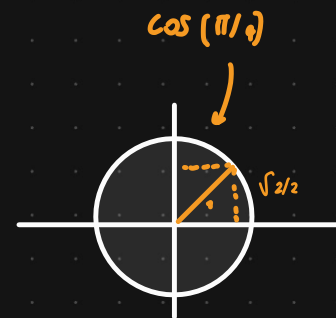
$$z_1 \cdot z_2 = \rho_1 e^{i\theta_1} \cdot \rho_2 e^{i\theta_2} = \rho_1 \rho_2 e^{i(\theta_1 + \theta_2)}$$

$$|z_1 \cdot z_2| = |z_1| |z_2|$$

$$\arg(z_1 \cdot z_2) = \arg(z_1) + \arg(z_2)$$

$$|y'\rangle = e^{i\theta} |y\rangle$$

$$\Rightarrow \langle y|A|y\rangle = \langle y'|A|y'\rangle$$



Given the state $|\phi\rangle = \frac{1}{2}|0_A\rangle|0_B\rangle + \frac{\sqrt{3}}{2}|0_A\rangle|1_B\rangle$, what matrix describes the subsystem A?

① $\rho_A = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{3}{2} \end{pmatrix}$

14% 3

② $\rho_A = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{\sqrt{3}}{2} \end{pmatrix}$

14% 3

③ $\rho_A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

43% 9

④ $\rho_A = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$

5% 1

⑤ $\rho_A = \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & \frac{3}{4} \end{pmatrix}$

24% 5

QUBIT: THINK OF A SPIN

$$\phi = \frac{\sqrt{3}}{2} |1\rangle_A |1\rangle_B + \frac{1}{2} |0\rangle_A |0\rangle_B \quad \rho_A = ?$$

IS ENTANGLED STATE,
CANNOT BE A PURE
STATE

ARE REDUNDANT

$$= \frac{\sqrt{3}}{2} |1\rangle_A |1\rangle_B + \frac{1}{2} |0\rangle_A |0\rangle_B$$

$$= \frac{\sqrt{3}}{2} |1\rangle_A |1\rangle_B + \frac{1}{2} |0\rangle_A |0\rangle_B$$

$$= \frac{\sqrt{3}}{2} |1\rangle_A |1\rangle_B + \frac{1}{2} |0\rangle_A |0\rangle_B$$

$$\rho_A = \text{Tr}_B(\rho)$$

$$\text{Tr}_B = |0\rangle_B \langle 0| + |1\rangle_B \langle 1|$$

$$\text{OR INSTEAD } \text{Tr}_B = \sum_n \langle n| \phi \otimes \phi | n \rangle_B =$$

$$= \langle 0| \rho | 0 \rangle_B + \langle 1| \rho | 1 \rangle_B$$

$$\rho = |\phi \otimes \phi\rangle = \left(\frac{\sqrt{3}}{2} |1\rangle_A |1\rangle_B + \frac{1}{2} |0\rangle_A |0\rangle_B \right) \left(\frac{\sqrt{3}}{2} \langle 1|_A \langle 1|_B + \frac{1}{2} \langle 0|_A \langle 0|_B \right)$$

$$= \frac{3}{4} |1\rangle_A |1\rangle_B \langle 1|_A \langle 1|_B + \frac{1}{4} |0\rangle_A |0\rangle_B \langle 0|_A \langle 0|_B + \frac{\sqrt{3}}{4} |1\rangle_A |1\rangle_B \langle 0|_A \langle 0|_B + \frac{1}{4} |0\rangle_A |0\rangle_B \langle 1|_A \langle 1|_B$$

THE MATRIX SHOULD HAVE SAME
VALUES ON SECOND DIAGONAL

$$\langle 0| \rho | 0 \rangle_B = \langle 0| \dots | 0 \rangle_B + \langle 1| \dots | 1 \rangle_B = \frac{1}{4} |0\rangle_A \langle 0|_A + \frac{3}{4} |1\rangle_A \langle 1|_A$$

MATRIX FORM

$$\text{Tr}_B(\rho) = \frac{1}{4} |0\rangle_A \langle 0|_A + \frac{3}{4} |1\rangle_A \langle 1|_A \quad \mathcal{B} = \{ |0\rangle_A, |1\rangle_A \}$$

$$\rho_{mn} = \langle m | \rho | n \rangle$$

$$\rho_{00} = 1/4$$

$$\rho_{11} = 3/4$$

$$\rho_{01} = 0$$

$$\rho_{10} = 0$$

$$\rho = \begin{pmatrix} 1/4 & 0 \\ 0 & 3/4 \end{pmatrix}$$

VERIFY ALSO THAT $\text{Tr}(\rho) = 1$

$$\frac{1}{4} + \frac{3}{4} = 1$$

$$|\psi\rangle = |\lambda_+\rangle$$

$$\mathcal{B} = \{|\lambda_+\rangle, |\lambda_-\rangle\}$$

$$\rho = |\psi\rangle\langle\psi| = |\lambda_+\rangle\langle\lambda_+|$$

$$\rho = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$|\lambda_+\rangle = \frac{1}{\sqrt{2}}(|2_+\rangle + |2_-\rangle)$$

$$\mathcal{B} = \{|2_+\rangle, |2_-\rangle\}$$

$$\rho = |\psi\rangle\langle\psi| = \frac{1}{2} \begin{bmatrix} |2_+\rangle\langle 2_+| + |2_+\rangle\langle 2_-| + |2_-\rangle\langle 2_+| + |2_-\rangle\langle 2_-| \end{bmatrix}$$

$$\rho = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

DIFFERENT BECAUSE
OF THE BASIS

What is the variance of the observable $O = 3|0\rangle\langle 0| - |1\rangle\langle 1| + (2+i)|0\rangle\langle 1| + (2-i)|1\rangle\langle 0|$ on the state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$?

1 $\langle \Delta O^2 \rangle = 5$ 17% 1

2 $\langle \Delta O^2 \rangle = 7$ 50% 3

3 $\langle \Delta O^2 \rangle = 0$ 33% 2

4 $\langle \Delta O^2 \rangle = 2$ 0% 0

5 $\langle \Delta O^2 \rangle = \sqrt{5}$ 0% 0

$$O = 3|0\rangle\langle 0| + (2+i)|0\rangle\langle 1| + (2-i)|1\rangle\langle 0| - |1\rangle\langle 1|$$

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$\langle \Delta O^2 \rangle?$$

$$\langle \Delta O^2 \rangle = \langle O^2 \rangle - \langle O \rangle^2$$

STEP 1: $\langle O \rangle$

option 1: $\mathcal{B} = \{|0\rangle, |1\rangle\}$ $O = \begin{bmatrix} \langle 0| & \langle 1| \\ 3 & 2+i \\ 2-i & -1 \end{bmatrix} \begin{pmatrix} |0\rangle \\ |1\rangle \end{pmatrix}$

$$\langle O \rangle = \frac{1}{2} (1 \ -1) \begin{pmatrix} 3 & 2+i \\ 2-i & -1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2} (1 \ -1) \begin{pmatrix} 1 & -i \\ 3 & -2 \end{pmatrix} \text{ FINIR } = -1$$

STEP 2: $\langle O^2 \rangle$

option 1
to compute O^2

$$O^2 = \begin{bmatrix} 3 & 2+i \\ 2-i & -1 \end{bmatrix} \begin{bmatrix} 3 & 2+i \\ 2-i & -1 \end{bmatrix} = \begin{bmatrix} 9+5 & 6+3i-2-i \\ 6-3i-2+i & 5+1 \end{bmatrix} = \begin{bmatrix} 14 & 4+2i \\ 4-2i & 6 \end{bmatrix} \text{ FINIR }$$

$$\langle O^2 \rangle = \frac{1}{\sqrt{2}} (1 \ -1) \begin{pmatrix} 14 & 4+2i \\ 4-2i & 6 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2} (1 \ -1) \begin{pmatrix} 14-4-2i \\ 4-2i-6 \end{pmatrix} = \frac{1}{2} (1 \ -1) \begin{pmatrix} 10-2i \\ -2-2i \end{pmatrix} = \dots = 6 \text{ FINIR }$$

OPTION 2 TO COMPUTE σ^2

$$\sigma^2 = \left(3|0x0\rangle + (2+i)|0x1\rangle + (2-i)|1x0\rangle - |1x1\rangle \right) \left(3|0x0\rangle + (2+i)|0x1\rangle + (2-i)|1x0\rangle - |1x1\rangle \right) =$$

= ... MORE COMPUTATIONS $\ddot{\wedge}$...

OPTION 3
TO COMPUTE
 σ^2

FIRST DIAGONALIZE σ ,
CHANGE THE BASIS $\ddot{\wedge}$

What are the possible outcomes of a measurement of the observable $M = \begin{pmatrix} 1 & i & 0 \\ -i & 2 & -i \\ 0 & i & 1 \end{pmatrix}$?

100

✓ ① {0, -1, 3} 0% 0

② {0, 0, 3} 27% 3

③ {0, 1, 3} 27% 3

④ {1, 1, 3} 9% 1

⑤ {1, 2, 3} 36% 4

A

λ_1

$|A - \lambda_1 I| = 0$ IF IS TRUE IT IS
AN EIGENVALUE

What is the representation of the state $|\psi\rangle = \sqrt{\frac{2}{3}}|x_+\rangle + \frac{1}{\sqrt{3}}|x_-\rangle$ in the basis $B = \{|z_+\rangle, |z_-\rangle\}$?

- a. $\begin{pmatrix} \sqrt{\frac{2}{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}$
- b. $\begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$
- c. $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$
- d. $\begin{pmatrix} \frac{\sqrt{2}+1}{\sqrt{6}} \\ \frac{\sqrt{2}-1}{\sqrt{6}} \end{pmatrix}$
- e. $\begin{pmatrix} \frac{2}{3} \\ \frac{1}{3} \end{pmatrix}$

$$|\psi\rangle = \sqrt{\frac{2}{3}}|x_+\rangle + \frac{1}{\sqrt{3}}|x_-\rangle \quad \text{on } B \{ |z_+\rangle, |z_-\rangle \}$$

$$|x_+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad |x_-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$|\psi\rangle = \sqrt{\frac{2}{3}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{\sqrt{3}} \left(\frac{1}{\sqrt{2}} \right) \begin{pmatrix} 1 \\ -1 \end{pmatrix} =$$

$$= \frac{1}{\sqrt{6}} \begin{pmatrix} \sqrt{2}+1 \\ \sqrt{2}-1 \end{pmatrix} \quad ??$$